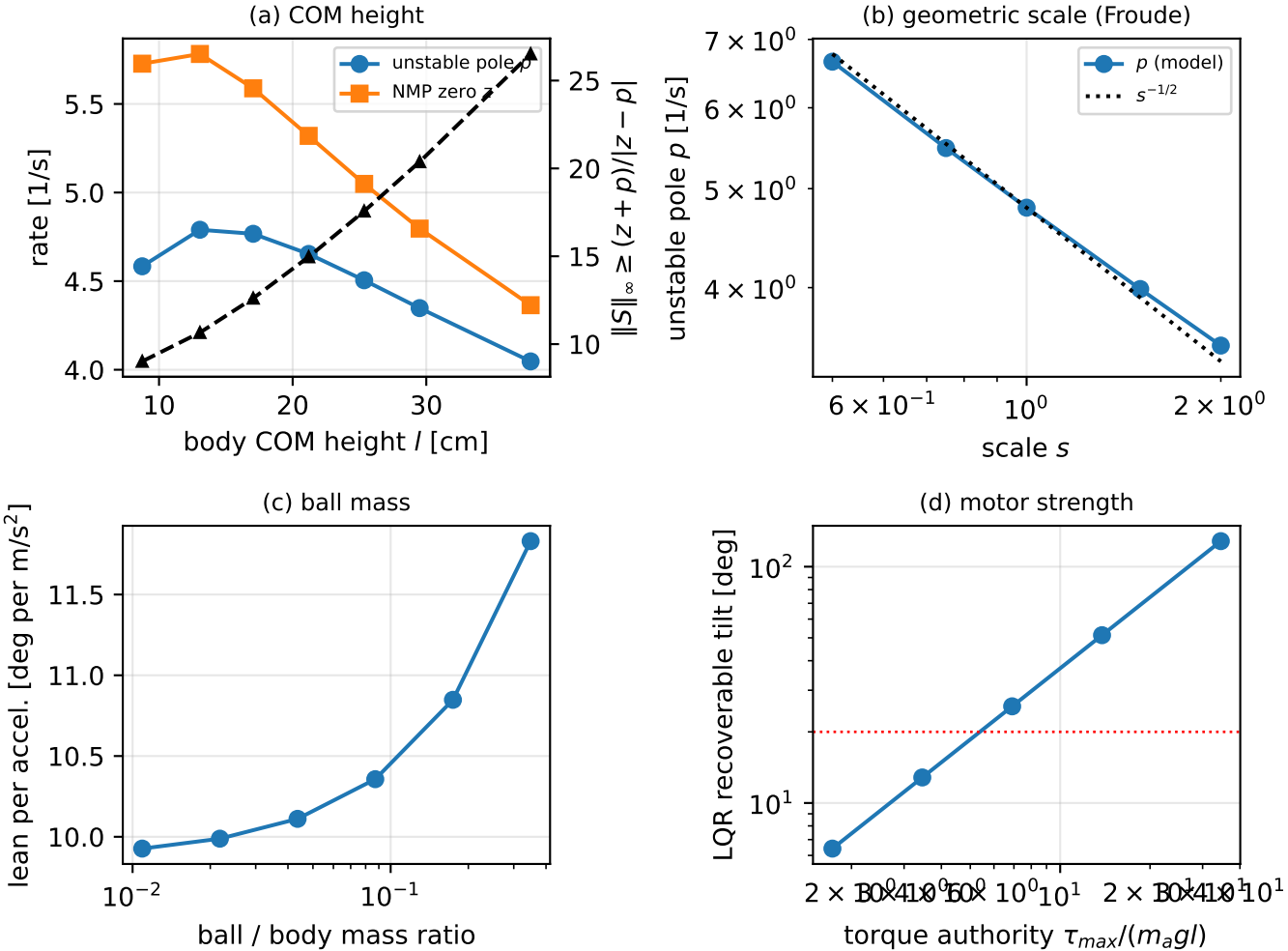
(a) COM height
unstable pole p
NMP zero z
rate [1/s]
$\|S\|_\infty \geq (z+p)/|z-p|$
body COM height l [cm]
(b) geometric scale (Froude)
p (model)
$s^{-1/2}$
unstable pole p [1/s]
scale s
(c) ball mass
lean per accel. [deg per m/s²]
ball / body mass ratio
(d) motor strength
LQR recoverable tilt [deg]
torque authority $\tau_{max}/(m_a g l)$